

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \mathcal{L}_S + \mathcal{L}_\chi + \mathcal{L}_T + \mathcal{L}_\psi + \mathcal{L}_{\text{cross}}$$

S : real scalar
 $\text{SU}(2)_L$ singlet, $Y = 0$

χ : Dirac fermion
 $\text{SU}(2)_L$ doublet, $Y = \frac{1}{2}$

T : complex scalar
 $\text{SU}(2)_L$ triplet, $Y = 1$

ψ : Majorana fermion
 $\text{SU}(2)_L$ singlet, $Y = 0$

Symmetries
 $\mathbb{Z}_2$ charges, $\text{U}(1)'$

Lagrangian Token Encoder

Field Token S

Field Token χ

Field Token T

Field Token ψ

Global Lagrangian
Token